GearDesign Report

Cylindrical gear(pair)

작성일: 2025년 10월 23일

Contents

1. Summary

1.1. Gear geometry

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal module (mm)	m_n	2.2500	
Pressure angle at normal section (deg)	α_n	20.0000	
Helix angle at reference circle (deg)	β	23.1000	
Manual normal backlash (mm)	j_bn	0.0000	
Number of teeth	z	22.0000	49.0000
Facewidth (mm)	b	36.0000	36.0000
Profile shift coefficient	x	0.0000	0.0000
Quality (ISO)	Q	6.0000	6.0000
Center distance (mm)	a	86.8380	
Tooth thickness tolerance	-	DIN 3967 cd25	DIN 3967 cd25
Gear material	-	DIN 18CrNiMo7 (σHlim = 430 Mpa)	DIN 18CrNiMo7 (σHlim = 430 Mpa)
Reference diameter (mm)	d	53.8150	119.8600
Tip diameter (mm)	d_a	58.315 (58.315 / 58.215)	124.360 (124.360 / 124.260)
Root diameter (mm)	d_f	48.190 (47.997 / 47.888)	114.235 (114.043 / 113.933)
Number of teeth spanned	k	4.0000	7.0000
Base tangent length (no backlash) (mm)	W_k	24.127 (24.062 / 24.024)	45.134 (45.068 / 45.030)
Effective Diameter of ball/pin (mm)	D_M	4.0000	4.0000
Diametral measurement over two balls (mm)	M_dk	59.623 (59.460 / 59.366)	125.691 (125.515 / 125.414)

1.2. Total safety factor

	Gear1 [Left]	Gear1 [Right]	Gear2 [Left]	Gear2 [Right]	Required S.F
Contact S.F	1.0825		1.1094		1.0000
Bending S.F	2.3191		2.4293		1.4000
Scuffing S.F (Flash Temp.)	2.2733				2.0000
Micropitting S.F	0.7424				1.0000

1.3. Total damage & life

	Gear1 [Left]	Gear1 [Right]	Gear2 [Left]	Gear2 [Right]
Contact Damage [%]	7.5000	0.0000	3.4000	0.0000
Bending Damage [%]	0.0000	0.0000	0.0000	0.0000
Contact Life [hr]	∞	∞	∞	∞
Bending Life [hr]	∞	∞	∞	∞
Total Req. Life [hr]	10000.0000			

1.4. Others

	Gear1 - Gear2
Transverse Contact Ratio	1.4254
Overlap Ratio	1.9982
Total Contact Ratio	3.4235
Worst Pitch Line Velocity [m/s]	28.1774
Total Efficiency [%]	98.9350
Worst Mesh Loss [W]	736.2355
Worst Churning Loss [W]	861.2311
Worst Peak-to-Peak T.E. [μm]	0.2823
Gear Total Mass [kg]	1.3186
Hunting tooth	Complete

1.5. Summary - Load cases

	Gear	Time [Hr]	Temperautre [ºC]	Power [kW]	Speed [r/min]	Torque [N.m]	Contact SF (Left)	Bending SF (Left)	Scuffing SF (Left)	Contact SF (Right)	Bending SF (Right)	Scuffing SF (Right)	K_Hβ (Left)	K_Hβ (Right)	Mesh Loss (Left) [kW]	Mesh Loss (Right) [kW]	Churning Loss [kW]	PPTE (Left) [μm]	PPTE (Right) [μm]
LC	Gear1	10000.0000	80.0000	150.0000	10000.0000	143.2390	1.0825	2.3191	2.2733				1.3000		0.7362		0.1108	0.2823	
1	Gear2			-150.0000	-4489.7960	319.0330	1.1094	2.4293									0.7504		

2. Detail geometry

2.1. Input gear geometry

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal module (mm)	m_n	2.2500	
Pressure angle at normal section (deg)	α_n	20.0000	
Helix angle at reference circle (deg)	β	23.1000	
Manual normal backlash (mm)	j_bn	0.0000	
Number of teeth	z	22.0000	49.0000
Facewidth (mm)	b	36.0000	36.0000
Profile shift coefficient	x	0.0000	0.0000
Quality (ISO)	Q	6.0000	6.0000
Center distance (mm)	a	86.8375	
Tooth thickness tolerance	-	DIN 3967 cd25	DIN 3967 cd25
Addendum coefficient	h*_aP	1.0000	1.0000
Dedendum coefficient	h*_fP	1.2500	1.2500
Root radius coefficient	ρ*_fP	0.3800	0.3800
Tip form height coefficient	h*_FaP	0.9556	0.9556
Ramp angle (deg)	α_KP	50.0000	50.0000
Protuberance height coefficient	h*_prP	0.0000	0.0000
Protuberance angle (deg)	α_prP	0.0000	0.0000
Tip alteration (mm)	k.m_n	0.0000	0.0000
Tip radius coefficient (for topping)	ρ*_aP	0.0000	0.0000
Gear material	-	DIN 18CrNiMo7 (σHlim = 430 Mpa)	DIN 18CrNiMo7 (σHlim = 430 Mpa)
Young's modulus (N/mm^2)	E	2.0600×10^5	2.0600×10^5
Poisson's ratio	ν	0.3000	0.3000
Density (kg/m^3)	ρ	7830.0000	7830.0000
Coeff. of thermal expansion (10^-8/ºC)	α_coeff	11.5000	11.5000
Surface hardness	HRC	61.0000	61.0000
Core hardness	HBW	325.0000	325.0000
Tensile strength (N/mm^2)	σ_B	1200.0000	1200.0000
Yield strength (N/mm^2)	σ_s	850.0000	850.0000
Specific heat capacity (J/kg.K)	c_M	485.0000	485.0000
Specific heat conductivity (W/m.K)	λ_M	50.0000	50.0000
Mean roughness height, root (um)	R_ZF	20.0000	20.0000
Mean roughness height, flank (um)	R_ZH	4.8000	4.8000
Roughness average value, root (um)	R_AF	3.0000	3.0000
Roughness average value, flank (um)	R_AH	0.6000	0.6000

	Symbol	Pair1 Gear1	Pair1 Gear2
Fatigue strength for bending stress (N/mm^2)	σ_Flim	430.0000	430.0000
Fatigue strength for contact stress (N/mm^2)	σ_Hlim	1500.0000	1500.0000

2.2. Calculated geometry - Gear

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Gear ratio	GR	2.2273	
Transverse module (mm)	m_t	2.4461	
Working normal pressure angle (deg)	α_wn	20.0000	
Transverse pressure angle (deg)	α_t	21.5885	
Working transverse pressure angle (deg)	α_wt	21.5885	
Working transverse pressure angle, e/i (deg)	α_wt, e/i	21.6069 / 21.5702	
Base helix angle (deg)	β_b	21.6340	
Helix angle at operating pitch circle (deg)	β_w	23.1000	
Radial backlash without tolerance (mm)	j_r0	0.0000	
Circumferential backlash without tolerance (mm)	j_t0	0.0000	
Min/max circumferential backlash by flank tolerance for each gear (mm)	j_t, e/i	0.0761 / 0.1196	0.0761 / 0.1196
Min/max normal backlash by flank tolerance for each gear (mm)	j_n, e/i	0.0658 / 0.1034	0.0658 / 0.1034
Circumferential backlash, e/i (mm)	j_tw, e/i	0.1435 / 0.2479	
Normal backlash, e/i (mm)	j_nw, e/i	0.1240 / 0.2143	
Angluar backlash, min (deg)	j_θ, min	0.3056	0.1372
Angluar backlash, max (deg)	j_θ, max	0.5278	0.2370
Effective facewidth (mm)	b_eff	36.0000	
Sum of the profile shift coefficients	Sum_x	0.0000	
Generating profile shift coefficient	x_E, e/i	-0.0427 / -0.0672	-0.0427 / -0.0672
Prevent undercut profile shift coefficient	x_Emin	-0.6190	-2.6059
Max. profile shift coeff. (min top land)	x_Emax	1.2629	2.1618
Reference center distance, x1=x2=0 (mm)	a_d	86.8375	
Working center distance (mm)	a_w	86.8375	
Reference diameter (mm)	d	53.8148	119.8602
Operating pitch diameter (mm)	d_w	53.8148	119.8602
Operating pitch diameter with tolerance (mm)	d_w, e/i	53.8216 / 53.8080	119.8754 / 119.8450
27		58.3148	124.3602
Tip diameter with tolerance (mm)	d_a, e/i	58.3148 / 58.2148	124.3602 / 124.2602

	Symbol	Pair1 Gear1	Pair1 Gear2
Root diamter (mm)	d_f	48.1898	114.2352
Root diameter with tolerance (mm)	d_f, e/i	47.9975 / 47.8876	114.0429 / 113.9330
Tip form diameter (mm)	d_Fa	58.1148	124.1602
Tip form diameter with tolerance (mm)	d_Fa, e/i	57.8996 / 57.7771	123.9563 / 123.8400
Root form diamter (mm)	d_Ff	50.6091	115.9195
Root form diameter with tolerance (mm)	d_Ff, e/i	50.5336 / 50.4928	115.7769 / 115.6964
Active tip diameter (EAP) (mm)	d_Na	58.1148	124.1602
Active tip diameter with tolerance (mm)	d_Na, e/i	57.8996 / 57.7771	123.9563 / 123.8400
Active root diameter (SAP) (mm)	d_Nf	50.8752	116.6251
Active root diameter with tolerance (mm)	d_Nf, e/i	51.0238 / 50.9498	116.8421 / 116.7333
Mean diameter for mass calculation (mm)	d_m	53.1037	119.1491
Gear mass (kg)	m	0.3587	0.9599
Gear inertia (kg.m^2)	J	0.0000	0.0000
Nominal tip clearance (mm)	c	0.5625	0.5625
Effective tip clearance, e/i (mm)	c, e/i	0.7746 / 0.6477	0.7746 / 0.6477
Min. margin of involute contact, (d_Nf_i - d_Ff_e) / 2 (mm)	c_NfFf	0.2081	0.4782

2.3. Calculated geometry - Tooth

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Tooth depth (mm)	h	5.0625	5.0625
Tooth depth, e/i (mm)	h, e/i	5.2136 / 5.1087	5.2136 / 5.1087
Working depth (mm)	h_w	4.5000	
Addendum (mm)	h_a	2.2500	2.2500
Addendum, e/i (mm)	h_a, e/i	2.2500 / 2.2000	2.2500 / 2.2000
Dedendum (mm)	h_f	2.8125	2.8125
Dedendum, e/i (mm)	h_f, e/i	2.9087 / 2.9636	2.9087 / 2.9636
Number of virtual gear teeth	z_n	27.6800	61.6509
Tooth thickness coeff.	s_nc	1.5708	1.5708
Transverse tooth thickness (mm)	s_t	3.8424	3.8424
Normal tooth thickness (mm)	s_n	3.5343	3.5343
Normal tooth thickness at d_a (mm)	s_an	1.6487	1.7737
Normal tooth thickness at d_a, e/i (mm)	s_an, e/i	1.6257 / 1.5311	1.7456 / 1.6603
Normal tooth thickness at d_Fa (mm)	s_Fan	1.7518	1.8615
Normal tooth thickness at d_Fa, e/i (mm)	s_Fan, e/i	1.8479 / 1.7440	1.9282 / 1.8369
Normal space width at d_f (mm)	e_fn	0.0000	1.7811
Normal space width at d_f, e/i (mm)	e_fn, e/i	0.0000 / 0.0000	1.8063 / 1.8213

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal pitch (mm)	p_n	7.0686	
Transverse pitch (mm)	p_t	7.6847	
Normal base pitch (mm)	p_bn	6.6423	
Transverse base pitch (mm)	p_bt	7.1457	
Axial pitch (mm)	p_x	18.0166	
Lead (mm)	p_z	396.3653	882.8137

2.4. Calculated geometry - Mesh

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Length of the path of contact (A to E) (mm)	g_α	10.1851	
Length of path of contact, e/i (mm)	g_α, e/i	9.7702 / 9.4551	
Length of approach POC of pinion (A to C) (mm)	g_f1	5.3091	
Length of approach POC of pinion, e/i (mm)	g_f1, e/i	5.0975 / 5.0563	
Length of recess POC of pinion (C to E) (mm)	g_a1	4.8761	
Length of recess POC of pinion, e/i (mm)	g_a1, e/i	4.6726 / 4.6541	
Length T1 - A (mm)	T1A	4.5912	
T1A tolerance (mm)	e/m/i	4.9864 / 4.8899 / 4.7935	
Length T1 - B (mm)	T1B	7.6307	
T1B tolerance (mm)	e/m/i	7.4180 / 7.3569 / 7.2958	
Length T1 - C (mm)	T1C	9.9003	
T1C tolerance (mm)	e/m/i	9.9095 / 9.9003 / 9.8910	
Length T1 - D (mm)	T1D	11.7369	
T1D tolerance (mm)	e/m/i	12.1320 / 12.0356 / 11.9391	
Length T1 - E (mm)	T1E	14.7763	
T1E tolerance (mm)	e/m/i	14.5636 / 14.5025 / 14.4414	
Length T2 - A (mm)	T2A	27.3597	
T2A tolerance (mm)	e/m/i	27.1275 / 27.0609 / 26.9944	
Length T2 - B (mm)	T2B	24.3202	
T2B tolerance (mm)	e/m/i	24.6850 / 24.5940 / 24.5030	
Length T2 - C (mm)	T2C	22.0506	
T2C tolerance (mm)	e/m/i	22.0712 / 22.0506 / 22.0300	
Length T2 - D (mm)	T2D	20.2140	
T2D tolerance (mm)	e/m/i	19.9818 / 19.9153 / 19.8487	
Length T2 - E (mm)	T2E	17.1745	
T2E tolerance (mm)	e/m/i	17.5393 / 17.4483 / 17.3573	
Length T1 - T2 (mm)	T1T2	31.9509	

	Symbol	Pair1 Gear1	Pair1 Gear2
T1T2 tolerance (mm)	e/m/i	31.9807 / 31.9508 / 31.9210	
Diameter of single contact point A (mm)	d_A	50.8752	124.1602
Diameter of single contact point B (mm)	d_B	52.3152	121.6036
Diameter of single contact point C (mm)	d_C	53.8148	119.8602
Diameter of single contact point D (mm)	d_D	55.2719	118.5579
Diameter of single contact point E (mm)	d_E	58.1148	116.6251
Transverse contact ratio	ε_α	1.4254	
Overlap ratio	ε_β	1.9982	
Total contact ratio	ε_Y	3.4235	
Total contact ratio with tolerance	e/m/i	3.365 / 3.343 / 3.321	
Specific sliding at Tip	ζ_a	0.4782	0.6262
Specific sliding at Root	ζ_f	-1.6755	-0.9163

2.5. Calculated geometry - Accuracy

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Quality Standard	-	ISO 1328: 2013	ISO 1328: 2013
Single pitch deviation (μm)	f_pt	8.5000	8.5000
Base circle pitch deviation (μm)	f_pb	7.9000	7.9000
Sector pitch deviation over k/8 pitches (μm)	F_pk/8T	18.0000	18.0000
Profile form deviation (μm)	f_fα	9.0000	9.0000
Profile slope deviation (μm)	f_Hα	7.0000	7.0000
Total profile deviation (μm)	F_α	11.0000	11.0000
Helix form deviation (μm)	f_fβ	10.0000	11.0000
Helix slope deviation (μm)	f_Hβ	9.0000	9.5000
Total helix deviation (μm)	F_β	13.0000	15.0000
Cumulative pith deviation, total (μm)	F_p	25.0000	28.0000
Runout (μm)	F_r	22.0000	25.0000
Single flank composite, tooth-to-tooth (μm)	f_isT	8.5000	8.5000
Single flank composite, total (μm)	F_isT	33.5000	36.5000
Tooth-to-tooth radial composite deviation (μm)	f_idT	9.5000	9.5000
Total radial composite deviation (μm)	F_idT	31.0000	31.0000

2.6. Calculated geometry - Measurement

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Number of teeth spanned	k	4.0000	7.0000
Base tangent length (mm)	W_k	24.1274	45.1336
Base tangent length with tolerance (mm)	W_k, e/i	24.0616 / 24.0241	45.0678 / 45.0302
Tolerance of W_k (mm)	ΔW_k, e/i	-0.0658 / -0.1034	-0.0658 / -0.1034
Diameter of contact point (mm)	d_M	54.8038	119.0593
Theoretical diameter of ball/pin (mm)	D_M.theory	3.8406	3.7955
Effective Diameter of ball/pin (mm)	D_M.eff	4.0000	4.0000
Radial single ball measurement (mm)	M_rk	29.8116	62.8769
Radial single ball measurement with tolerance (mm)	M_rk, e/i	29.7300 / 29.6830	62.7887 / 62.7380
Diametral two ball measurement (mm)	M_dk	59.6231	125.6913
Diametral two ball measurement with tolerance (mm)	M_dk, e/i	59.4600 / 59.3659	125.5150 / 125.4136
Tolerance of M_dk (mm)	ΔM_dk	-0.1631 / -0.2572	-0.1764 / -0.2777

3. Detail rating

3.1. General

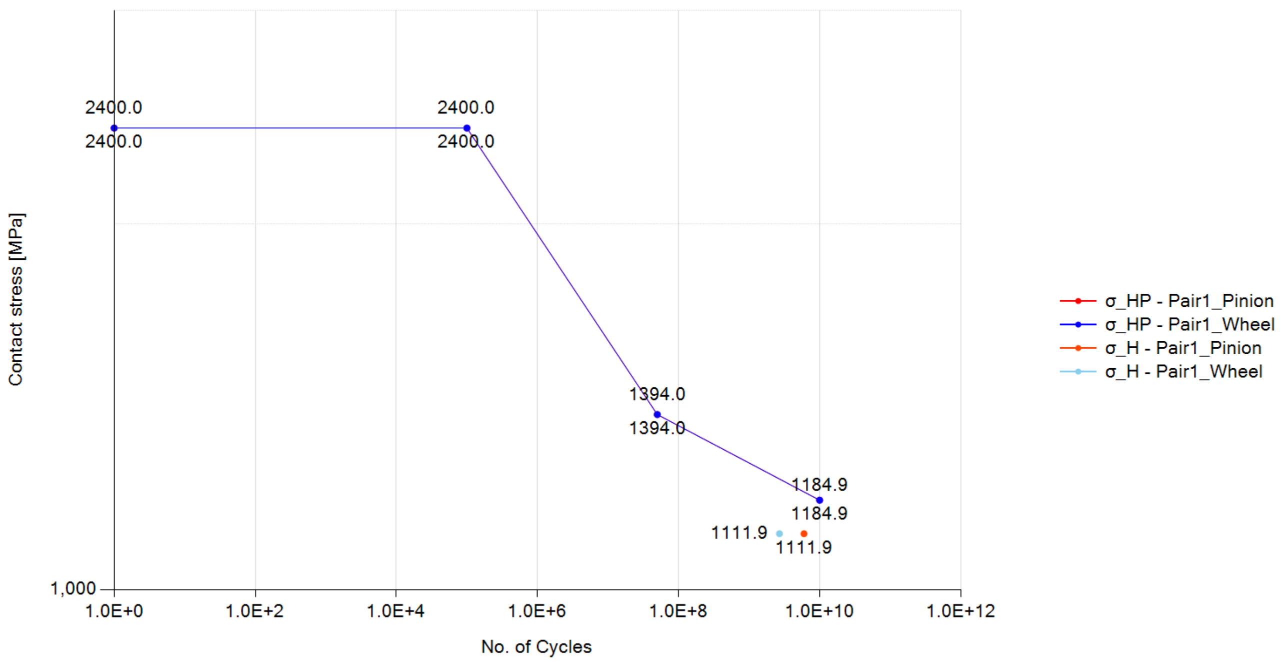
Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rotational speed (rpm)	ω	10000.0000	4489.7960
Nominal tangential load at d (N)	F_t	5323.4230	
Nominal axial load at d (N)	F_a	2270.6320	
Nominal radial load at d (N)	F_r	2106.4600	
Nominal normal load at d (N)	F_n	6158.8780	
Nominal tangential load per unit facewidth (N/mm)	w	147.8730	
Circumferential speed at d (m/s)	v	28.1770	
Sliding velocity at Tip dia. (m/s)	v_g	7.3990	8.0560
Nominal tangential load at d_w (N)	F_tw	5323.4230	
Nominal axial load at d_w (N)	F_aw	2270.6320	
Nominal radial load at d_w (N)	F_rw	2106.4600	
Circumferential speed at d_w (m/s)	v_w	28.1770	
Running-in value (um)	y_p	0.5930	
Running-in value (um)	y_f	0.6750	
Effective single pitch & profile deviation (um)	f_pb_eff	7.3080	
Effective single pitch & profile deviation (um)	f_fa_eff	8.3250	
Correction factor	C_M	0.8000	
Gear blank factor	C_R	0.8980	
Basic rack factor	C_B	0.9750	
Material factor	E/E_st	1.0000	
Theoretical single stiffness (N/mm/um)	c'_th	17.5340	
Single tooth stiffness (N/mm/um)	c'	11.2970	
Mesh stiffness (for Kv, KHα, KFα), (N/mm/um)	c_yα	14.9010	
Mesh stiffness (for KHβ, KFβ), (N/mm/um)	c_yβ	12.6660	
Mean diameter for K_v (mm)	d_m (K_V)	53.2520	119.2980
Moment of inertia for K_v (kg.mm^2)	J (K_V)	210.4910	3645.4340
Moment of gear inertia per unit faccewidth (kg.mm)	J*	5.8470	101.2620
Relative gear mass per unit facewidth (kg/mm)	m*	0.0093	0.0326
Relative gear mass per unit facewidth (kg/mm)	m_red	0.0073	
Resonance running speed (rpm)	n_E1	19663.8100	
Resonance ratio	N	0.5090	
Resonance ratio in main resonance ratio	N_s	0.8500	
Resonance range	-	subcritical range	
Dynamic factor	K_V	1.1460	

	Symbol	Pair1 Gear1	Pair1 Gear2
Face load factor - flank	K_Hβ	1.3000	
Face load factor - root	K_Fβ	1.2530	
Transverse load factor - flank	K_Hα	1.1170	
Transverse load factor - root	K_Fα	1.1170	
Number of load cycles	N_L	6.0000×10^9	2.6940×10^9

3.2. Pitting

S-N curve (Pitting), Load case 1, Meshing pair: Gear1 - Gear2



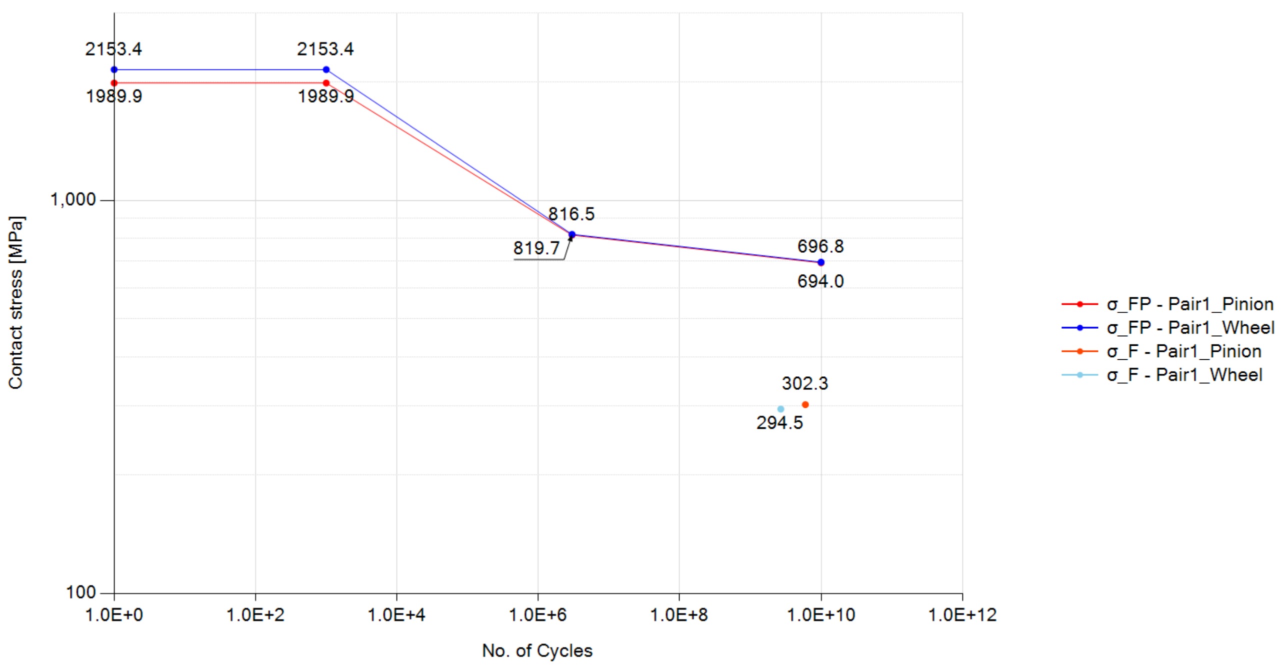
Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rating Method	-	ISO 6336-2:2019	
Contact ratio factor	Z_ε	0.8380	
Zone factor	Z_H	2.3310	
Elasticity factor	Z_E	189.8120	
Young's modulus for tooth flank (N/mm^2)	E	2.0600×10^5	2.0600×10^5
Helix angle factor	Z_β	1.0430	
Single pair tooth contact factor (pinion)	Z_B	1.0000	
Single pair tooth contact factor (wheel)	Z_D	1.0000	
Nominal contact stress at the pitch point (N/mm^2)	σ_H0	771.0680	
Contact stress (N/mm^2)	σ_H	1111.8890	1111.8890
Life factor	Z_NT	0.8630	0.8850

	Symbol	Pair1 Gear1	Pair1 Gear2
Lubricant factor	Z_L	0.9460	0.9460
Velocity factor	Z_V	1.0310	1.0310
Roughness factor	Z_R	0.9530	0.9530
Work hardening factor	Z_W	1.0000	1.0000
Size factor	Z_X	1.0000	1.0000
Permissible contact stress - static (N/mm^2)	σ_HP_static	2400.0000	2400.0000
Permissible contact stress - mid point (N/mm^2)	σ_HP_mid	0.0000	0.0000
Permissible contact stress - reference (N/mm^2)	σ_HP_ref	1393.9800	1393.9800
Permissible contact stress - long (N/mm^2)	σ_HP_long	1184.8830	1184.8830
Permissible contact stress (N/mm^2)	σ_HP	1203.5950	1233.5250
Safety factor for Contact	S_H	1.0820	1.1090
Contact damage (%)	-	7.5490	3.3890
Cycle to failure	-	7.9480×10^10	7.9480×10^10

3.3. Bending

S-N curve (Bending), Load case 1, Meshing pair: Gear1 - Gear2



Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rating Method	-	ISO 6336-3:2019	
Tooth form factor	Y_F	1.2080	1.0750
Stress correction factor	Y_S	1.8310	2.0040

	Symbol	Pair1 Gear1	Pair1 Gear2
Working angle (deg)	α_Fen	18.6930	19.8320
Bending moment arm (mm)	h_Fe	2.3400	2.4730
Tooth thickness at root (mm)	s_Fn	4.5300	4.9180
Tooth root radius (mm)	ρ_F	1.2980	1.1600
Helix angle factor	Y_β	1.0380	
Deep tooth factor	Y_DT	1.0000	
Rim thickness factor	Y_B	1.0000	1.0000
Nominal tooth root stress (N/mm^2)	σ_F0	150.8730	146.9400
Tooth root stress (N/mm^2)	σ_F	302.3450	294.4630
Life factor	Y_NT	0.8590	0.8730
Relative notch sensitive factor	Y_δrelT	0.9920	0.9960
Surface factor	Y_RrelT	0.9570	0.9570
Size factor	Y_X	1.0000	1.0000
Stress correction factor	Y_ST	2.0000	
Mean stress influence factor	Y_M	1.0000	1.0000
Technical factor	Y_T	1.0000	1.0000
Permissible tooth root stress - static (N/mm^2)	σ_FP_static	1989.9260	2153.3920
Permissible tooth root stress - reference (N/mm^2)	σ_FP_ref	816.5070	819.7410
Permissible tooth root stress - long (N/mm^2)	σ_FP_long	694.0310	696.7800
Permissible tooth root stress (N/mm^2)	σ_FP	701.1700	715.3330
Safety factor for Bending	S_F	2.3190	2.4290
Bending damage (%)	-	0.0000	0.0000
Cycle to failure	-	5.2310×10^20	2.3820×10^21

3.4. Scuffing

Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Lubrication type	-	Spray lubrication	
Lubricant temperature (deg)	θ_oil	80.0000	
Lubricant viscosity (mPa.s)	η_oil	13.1510	
Load stage according to FZG A/8, 3/90	S_FZG	12.0000	
Gear density (kg/m^3)	ρ_M	7830.0000	7830.0000
Gear specific heat (J/kg.K)	c_M	485.0000	485.0000
Gear heat conductivity (N/s.K)	λ_M	50.0000	50.0000
Gear roughness (μm)	R_a	0.6000	0.6000
Number of mating gears	n_p	1.0000	
Effective facewidth (mm)	b_eff	30.6000	

	Symbol	Pair1 Gear1	Pair1 Gear2
Relevant tip relief (μm) (Min. value = 2 μm)	C_a	2.0000	2.0000
Transverse unit load (N/mm)	w_Bt	307.4860	
Lubrication system factor	X_S	1.2000	
Multiple meshing factor	X_mp	1.0000	
Structural factor	X_W	1.0000	1.0000
Structural factor (total)	X_Wtot	1.0000	
Thermal contact factor	B_M	13.7800	13.7800
Multiple path factor	K_mp	1.0000	
Optimal tip relief (μm)	C_eff	12.4050	
Angle factorr	X_αβ	0.9660	
Lubricant factor	X_L	0.8790	
Roughness factor	X_R	0.7750	
Mean coefficient of friction	μ_m	0.0480	
Maximum flash temperature (deg)	θ_fl_max	105.4560	
Mean flash temperature (deg)	θ_fl_mean	35.1790	
Total mass temperature (deg)	θ_M	99.8410	
Scuffing temperature (deg)	θ_S	364.8370	
Maximum contact temperature (deg)	θ_Bmax	205.2970	
Safety factor for flash temperature	S_B	2.2730	

3.5. Micro-pitting

Load case 1

	Symbol	Gear1 - Gear2
minimum lubricant film thickness (μm)	h_Ymin	0.0940
minimum specific lubricant film thickness in the contact area	λ_GFmin	0.1567
permissible specific lubricant film thickness	λ_GFP	0.2111
safety factor against micropitting	S_λ	0.7424
transverse tangential load at reference cylinder per mesh (N)	F_t	5323.4230
transverse load in plane of action (N)	F_bt	5725.0340
reduced modulus of elasticity (N/mm^2)	E_r	2.2637×10^5
thermal contact coefficient of pinion (N/(ms^0.5*K))	B_M1	13779.6040
thermal contact coefficient of wheel (N/(ms^0.5*K))	B_M2	13779.6040
failure load stage according to FVA 54/7 (90º)	-	0.0000
effective arithmetic mean roughness value (μm)	R_a	0.6000
roughness factor	X_R	1.1760
lubricant factor	X_L	1.0000
lubrication factor	X_S	1.2000

	Symbol	Gear1 - Gear2
elasticity factor (N/mm^2)^0.5	Z_E	189.8120
helical load factor	K_Bγ	1.3000
load losses factor	H_v	0.1320
mean coefficient of friction	μ_m	0.0570
profile position at min. lubricant film thickness	-	0.0000
tip relief factor	X_Ca	1.0000
load sharing factor	X_A	0.9120
buttressing factor	X_but,A	1.3000
effective tip relief (μm)	C_eff	0.0000
Applied tip relief in calculation (μm)	C_a	0.0000
sum of tangential velocities at point A (m/s)	v_Σ,A	17.6720
sliding velocity at point A (m/s)	v_g,A	-8.0560
tangential velocity on pinion at point A (m/s)	v_r1,A	4.8080
tangential velocity on wheel at point A (m/s)	v_r2,A	12.8640
oil inlet/sump temperature (ºC)	θ_oil	80.0000
bluk temperature (ºC)	θ_M	104.6840
flash temperature at point A (ºC)	θ_fl,A	127.6440
contact temperature at point A (ºC)	θ_B,A	232.3270
pressure-viscosity coefficient at 38 ºC (m^2/N)	α_38	1.8390×10^-8
pressure-viscosity coefficient at bulk temperature (m^2/N)	α_θM	1.3000×10^-8
pressure-viscosity coefficient at point A contact temperature (m^2/N)	α_θB,A	6.6570×10^-9
kinematic viscosity at 40 ºC (mm^2/s)	ν_40	68.0000
kinematic viscosity at 100 ºC (mm^2/s)	ν_100	8.8000
kinematic viscosity at point A contact temperature (mm^2/s)	ν_θB,A	1.4290
dynamic viscosity at 38 ºC (Ns/m^2)	η_38	0.0650
dynamic viscosity at bulk temperature (Ns/m^2)	η_θM	0.0070
dynamic viscosity at oil inlet/sump temperature (Ns/m^2)	η_θoil	0.0120
dynamic viscosity at point A contact temperature (Ns/m^2)	η_θB,A	0.0010
normal radius of relative curvature at point A (mm)	ρ_n,A	4.2290
material parameter	G_M	2943.7350
local velocity parameter at point A	U_A	5.9990×10^-11
local load parameter at point A	W_A	0.0003
local sliding parameter at point A	S_GF,A	0.0820
local nominal Hertzian contact stress at point A (Method B) (N/mm^2)	p_H,A	1111.5560
local Hertzian contact stress at point A including the load factors K (Method B) (N/mm^2)	p_dyn,A	1602.8750

4. Power loss

4.1. Mesh loss

Load case 1

	Symbol	Gear1 - Gear2
Rating Method	-	ISO/TR 14179-2:2001
Input Power (kW)	P	150.0000
Gear ratio	u	2.2273
Tangential speed at working pitch dia. (m/s)	v_wt	28.1774
Tangential force at pitch dia. (N)	F_t	5323.4232
Tooth normal force, transverse section (N)	F_bt	5725.0336
Arithmetic average roughness of pinion	Ra1	0.6000
Arithmetic average roughness of wheel	Ra2	0.6000
Roughness factor	X_R	0.6000
Sum velocity (m/s)	v_Σ	20.7351
Equivalent radius of curvature (mm)	ρ	7.3504
Kinematic viscosity of oil at operating temp. (cSt)	η_oil	14.8602
Lubricant type	-	Mineral oil base
Lubricant factor	X_L	1.0000
distributed force on tooth surface(N/m)	F/b	159.0287
Mean coefficient of friction of the gear mesh	μ_mz	0.0372
Tooth loss factor	H_v	0.1318
Mesh power loss (W)	P_VZP	736.2355
Mesh efficiency (%)	η_z	99.5092

4.2. Churning loss

Load case 1

	Symbol	Gear1	Gear2
Rating Method	-	Changenet (2011)	Changenet (2011)
Rotating speed (rpm)	n	10000.0000	4489.7959
Lubricant	-	Oil: ISO-VG 68	Oil: ISO-VG 68
Gear immerged depth (mm)	h_e	26.9074	59.9301
Oil density (kg/m^3)	ρ	885.0000	885.0000
Operating Viscocity (cSt)	ν	14.8602	14.8602
dynamic viscosity (kg/m.s)	μ	0.0000	0.0000
Reynolds number	Re	68261.8772	68261.8772
Froude number	Fr	3007.8756	1350.4748
Centrifugal acceleration (m/s^2)	γ	14218.1402	3743.0055

	Symbol	Gear1	Gear2
Tip diameter of gear (mm)	d_a	58.3148	124.3602
Pitch diameter of gear (mm)	d_p	53.8148	119.8602
Facewidth (mm)	b	36.0000	36.0000
Dimensionless drag torque	C_m	0.0011	0.0027
Immersion area of the gear (m^2)	S_m	0.0100	0.0284
Total oil churning loss (W)	P	110.8149	750.4162
Total oil churning torque (N.m)	T	0.1058	1.5961

4.3. Windage loss

Load case 1

	Symbol	Gear1	Gear2
Rating Method	-	None	None
Windage loss (W)	P_WL	0.0000	0.0000
Windage torque (N.m)	T_WL	0.0000	0.0000